

On-Site Field Service Report

Uninterruptible Power Supply — Corrective & Preventive Maintenance

Report	Detail	Asset	Value
Report No.	FSR-2025-GRE-0417	Serial no.	GVS-0002
Service date	2025-04-17	Model	Galaxy VS (3-phase UPS)
Technician	A. Moreau (Cert. L2)	Site	Grenoble Data Center — SITE-GRE-02
Dispatch type	Corrective — fault alarm	Firmware	v2.4
Work order	WO-88213	Warranty	In warranty (exp. 2027-12-31)

1. Equipment Specifications

Parameter	Rating	Measured
Topology	Online double-conversion (VFI-SS-111)	—
Rated power	10–100 kVA / kW (pf = 1.0)	80 kVA
Nominal DC bus	480 V DC	511 V DC ■
Input THD	< 3% at full load	2.4%
Efficiency	up to 97% (double-conv.)	96.1%
Battery design life	5 years @ 25 °C	Year 3
PM interval	12 months	Due 2025-05

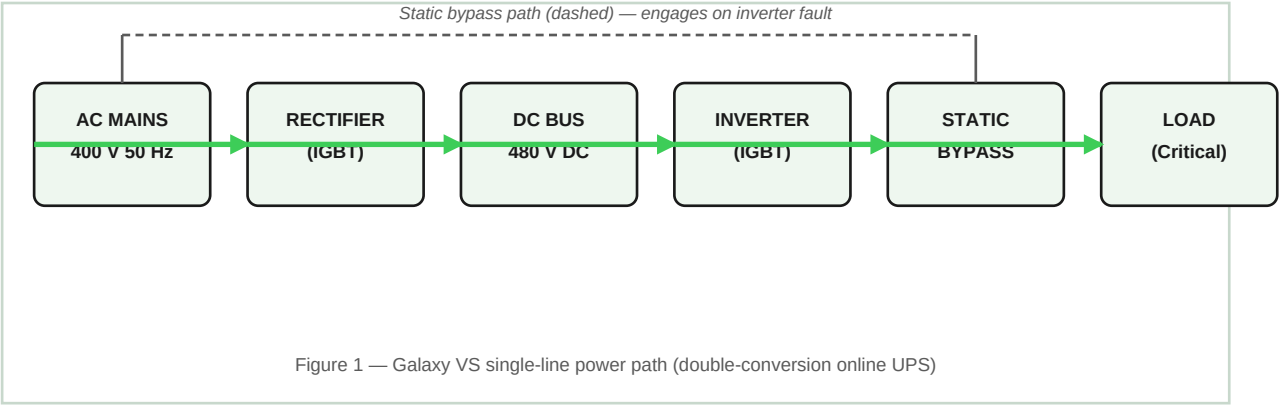
2. Fault Code Reference

Code	Meaning	First technician action
E07	DC bus overvoltage on the capacitor bank	Transfer load to static bypass, isolate the UPS, then inspect the DC bus and rectifier IGBT stage before restart.
E12	Battery string below end-of-discharge threshold	Verify battery breaker closed; measure per-block voltage; replace any block < 1.75 V/cell.
E21	Overtemperature on inverter heatsink	Check air filters and fans; confirm intake < 40 °C; clear obstruction, then reset.
E33	Static bypass out of tolerance (V/Hz)	Confirm source within ±10% V and ±1 Hz; do NOT force bypass while out of tolerance.

3. Power Termination Torque Specifications

Termination	Thread	Torque (N·m)	Torque (lb·in)	Re-torque
Input bus bar	M10	40	354	✓
Output bus bar	M10	40	354	✓
Battery cabinet link	M8	20	177	✓
Earth / PE stud	M8	20	177	✓
Control terminal	M3.5	0.8	7	—

4. Single-Line Power Path



5. Preventive Maintenance Checklist

Check	Pass	Fail	Note
Air filters & fan operation	■	■	Cleaned
Battery block voltages	■	■	All > 1.9 V/cell
DC bus voltage within tolerance	■	■	511 V — E07 raised
Torque audit of power terminations	■	■	Per section 3
Firmware baseline recorded	■	■	v2.4
Static bypass transfer test	■	■	Nominal

6. Technician Observations

During a routine preventive-maintenance visit the unit raised **fault code E07** (DC bus overvoltage). Measured DC bus was 511 V DC against a 480 V nominal. Load was transferred to **static bypass** and the UPS isolated per lockout-tagout (LOTO). The DC capacitor bank was verified to zero energy with a meter before contact (arc-flash PPE worn throughout). Inspection of the **rectifier IGBT stage** found no visible damage; a gate-driver anomaly is suspected. Recommend firmware review and a follow-up with the escalation engineering team for asset GVS-0002.

7. Warranty & Service Contract

Field	Value
Service plan	EcoStruxure Service Plan — Premium
Warranty status	In warranty
Coverage end	2027-12-31
Escalation candidate	Yes — recurring DC bus event

Synthetic workshop sample. This document was fabricated for a hands-on Microsoft Foundry training lab. Product names, serial numbers, sites, fault codes, torque values and warranty records are fictitious and are used only for realism. This content is **not affiliated with, produced by, or endorsed by Schneider Electric**.